

Blair Subbaraman

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Education

- 2020-Present PhD, Human Centered Design & Engineering, *University of Washington*, Seattle, WA
Anticipated Graduation: June 2026
Advisor: Nadya Peek
Committee: Mako Hill, Daniela Rosner, Casey Reas
- 2022 MS, Human Centered Design & Engineering, *University of Washington*, Seattle, WA
- 2018 BA, Physics, *Pomona College*, Claremont, CA
Advisor: Dwight Whitaker
Minor: Mathematics

Publications

Refereed Conference and Journal Articles

- 2025 **Blair Subbaraman**, Nathaneal Bursch, and Nadya Peek. It's Not the Shape, It's the Settings: Tools for Exploring, Documenting, and Sharing Physical Fabrication Parameters in 3D Printing. In Proceedings of the 2025 ACM CHI Conference on Human Factors in Computing Systems (CHI '25) 🏆 Best Paper Honorable Mention (Top 5% of all accepted papers)
- 2024 **Blair Subbaraman**, Orlando de Lange, Sam Ferguson, and Nadya Peek. The Duckbot: A System for Automated Imaging and Manipulation of Duckweed. In Plos one 19, no. 1 (2024)
- 2023 **Blair Subbaraman** and Nadya Peek. 3D Printers Don't Fix Themselves: How Maintenance is Part of Digital Fabrication. In Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23).
- 2023 **Blair Subbaraman**, Shenna Shim, and Nadya Peek. Forking a Sketch: How the OpenProcessing Community Uses Remixing to Collect, Annotate, Tune, and Extend Creative Code. In Proceedings of the 2023 ACM Designing Interactive Systems Conference (DIS '23).
- 2022 **Blair Subbaraman** and Nadya Peek. P5.fab: Direct Control of Digital Fabrication Machines from a Creative Coding Environment. In Proceedings of the 2022 ACM Designing Interactive Systems Conference (DIS '22).

- 2022 Xiaowen Sun, Daniel Gilman Calderón, **Blair Subbaraman**, and Jeffrey A. Burke. An Audience Data-Driven Alternate Reality Storytelling Design. In International Conference on Human-Computer Interaction (C&C '22).

Conference Posters and Juried Demonstrations

- 2024 **Blair Subbaraman** and Nadya Peek. Playing the Print: MIDI-based Fabrication Interfaces to Explore and Document Material Behavior. In Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems (CHI EA '24).
- 2022 **Blair Subbaraman** and Nadya Peek. Demonstrating p5.fab: Direct Control of Digital Fabrication Machines From a Creative Coding Environment. In Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct).
- 2021 **Blair Subbaraman** and Nadya Peek. Demonstrating p5.fab: Direct Control of Digital Fabrication Machines From a Creative Coding Environment. In ACM Symposium on Computational Fabrication (SCF '21).
- 2021 Gabrielle Benabdallah and **Blair Subbaraman**. BodyTeller: Explorations in Narrative Biosensing, In Society for Social Studies of Science (4S '21).

Other Writing

- 2025 **Blair Subbaraman**. Printed Patterns. Algorithmic Pattern Conference (Alpaca 2025). Sheffield, UK. 2022.
- 2025 Brenden Pelkie et al. (including **Blair Subbaraman**). Democratizing Self-driving Labs Through User-developed Automation Infrastructure. ChemRxiv. 2025.
- 2022 Gabrielle Benabdallah and **Blair Subbaraman**. Explorations in narrative biosensing. Interactions. ACM New York, NY, USA, 2022.

Professional Activities

Talks

** Indicates invited talk*

- 2025 Printed Patterns, *Algorithmic Pattern Conference*, Sheffield, England.

- 2025 Automation for Insight, *NC State Building Future Faculty Program*, Raleigh, NC, USA.
- 2024 Sketching with Machines, *Hackaday Superconference*, Pasadena, CA, USA.
- 2024* Science Jubilee: Open-source Hardware for Laboratory Automation, *Duckweeds, Microbes, and Symbiosis*, University of British Columbia, Vancouver, Canada.
- 2023* Computational Fabrication in Performance Contexts, *UCLA School of Theater, Film, & Television*, Los Angeles, CA, USA.
- 2023 Bridging Creative Code and Digital Fabrication, *Open Source Arts Contributors Conference*, Clinic for Open Source Arts (COSA), University of Denver, CO, USA.
- 2023 Custom Workflow Automation with Jubilee, *Global Community Biosummit 6.0*, Virtual & MIT
- 2022* Niche Experiments with Jubilee: Duckweed Growth Assays as a Case Study, *Duckweeds and Microbes: A New Frontier in Symbiosis Research*, University of Toronto.

Workshops

- 2024, 2025 Pathways to Open-Source Hardware for Laboratory Automation, *University of Washington*, Seattle, Washington, USA, Co-Organizer.
Co-organized two consecutive NSF-sponsored three-day workshop with 25 participants. The workshops gathered an international group of scientists and engineers using open-source technologies for automating scientific experiments to build and share their approaches.
- 2025 3D Printing with p5.fab, *Creative Coding Fest*, Online
Organized multiple online workshops with over 20 educators and hobbyists to introduce 3D printing techniques using Javascript code.
- 2021 Creative Fabrication Workshop, *University of Washington*
Workshop with professional artists and seasoned makers to explore the possibilities of controlling machines with code.

Awards and Honors

- 2025 CHI Best Paper Honorable Mention

- 2024 Processing Foundation Fellowship Finalist
- 2018 Richard P. Edmunds Physics Prize for outstanding graduating student, Pomona College

Teaching Experience

- 2025 Instructor of Record, HCDE 439: Physical Computing, *University of Washington*
Undergraduate level introduction to engineering and prototyping interactive systems & environments for human-centered applications. 40 students.
- 2021 Graduate Teaching Assistant, HCDE 439: Physical Computing, *University of Washington*
Sole TA, 25 students.
- 2018 Undergraduate Teaching Assistant, PHYS 128: Electronics with Laboratory, *Pomona College*
Transistors and integrated circuits in a variety of applications including operational amplifiers, basic digital circuits, analog/digital conversion and an introduction to microprocessors. 2 TAs, 30 students.
- 2017 Undergraduate Teaching Assistant, PHYS 41: General Physics with Laboratory, *Pomona College*
Calculus-based introduction to Newtonian mechanics and thermodynamics for non-majors. 2 TAs, 25 students.
- 2015, 2016 Undergraduate Teaching Assistant, PHYS 70: Spacetime, Quanta and Entropy with Laboratory, *Pomona College*
Calculus-based introduction to principles of contemporary physics. Topics include conservation laws, special relativity, quantum physics and thermal physics, all viewed from a 21st century perspective. 2 TAs, 30 students.

Mentorship

* Signifies co-authorship on peer-reviewed article

- 2025- Catherine Zhang, Bachelor's Student, Computer Science, University of
Present Washington.
- 2024- Nathaneal Bursch*, Bachelor's Student, Mechanical Engineering, University
2025 of Washington.

- 2022-2024 Sam Ferguson*, Bachelor's Student, Human Centered Design and Engineering, University of Washington.
- 2022-2023 Shenna Shim*, Bachelor's Student, Human Centered Design and Engineering, University of Washington.
- 2021-2022 Annie Liu, Bachelor's Student, Human Centered Design and Engineering, University of Washington.
- 2021-2022 Kenny Le, Bachelor's Student, Human Centered Design and Engineering, University of Washington.

Service

Reviewing

** Special Recognition for Outstanding Reviews*

- 2025 ACM Conference on Human Factors in Computing (CHI)*
ACM Conference on Designing Interactive Systems (DIS)*
ACM Symposium on User Interface Software and Technology (UIST)*
- 2024 ACM Conference on Human Factors in Computing (CHI)*
ACM Conference on Designing Interactive Systems (DIS)*
Halfway to the Future Symposium (HttF)
- 2023 ACM Conference on Human Factors in Computing (CHI)*
ACM Conference on Designing Interactive Systems (DIS)*

Conference Service

- 2024 Student Volunteer, ACM Conference on Human Factors in Computing (CHI)
- 2022 Session Chair, ACM Symposium on Computational Fabrication (SCF)
Student Volunteer, Eyeo Festival
- 2017 Student Volunteer, Eyeo Festival

Research Positions

- 2020- Present Machine Agency
with Nadya Peek, University of Washington
Topics: Digital and computational fabrication, laboratory automation
- 2022 Slip Rabbit Studio
with Timea Tihanyi & Audrey Desjardins, University of Washington
Topics: Data physicalization, ceramic 3D printing
- 2021 Tactile and Tactical Design Lab
with Daniela Rosner, University of Washington
Topics: Biosensing, wearable technologies
- 2018- 2020 Center for Research in Engineering, Media, & Performance (REMAP)
with Jeff Burke, UCLA
Topics: AR theater performance, open source software for the arts
- 2014- 2018 Fluids Dynamics Lab
with Dwight Whitaker, Pomona College
Topics: Biological hydrodynamics, scientific instrumentation

Performances & Exhibitions

- 2020 The Invention of Morel, Virtual Performance
Developer for a workshop and performance exploring distributed remote virtual production during COVID-19 through an adaptation of The Invention of Morel by Adolfo Bioy Casares.
- 2019- 2020 The Man in the High Castle, UCLA Freud Playhouse & Virtual
AR app development and HCI research for an experimental, immersive theater piece based on the world of the streaming series The Man in the High Castle.
- 2019 @LAs, UCLA Freud Playhouse
Developer and designer working with directors, cinematographers, and set designers to realize an AI/ML driven pop-up exhibition prototype as part of UCLA's Future Storytelling Summer Institute 2019.
- 2018 The Buzz, UCLA Campus
Created and deployed the sound platform for a Raspberry Pi based audio/furniture installation in collaboration with UCLA cityLAB.
- 2018 Entropy Bound, UCLA Little Theater
Developed software for custom wearables to support an experimental play weaving dramaturgy with code.

Selected Synergistic Activities

2021- Present	p5.fab Developed and maintain p5.js library for authoring physical artifacts.
2023- Present	Science Jubilee Developed and maintain open-source software and hardware for laboratory automation.
2023- 2025	p5.js Math Steward One of ten stewards for issues, contributions, and community discussion related to the Math features of the open source creative coding library p5.js.
2018- 2021	OpenPTrack Community Manager Managed issues, questions, and discussions for OpenPTrack, an open source person tracking software for the arts.
2016- 2018	Community Science Projects Led partnership between Pomona College and Fremont Academy of Engineering & Design, a public high school in Pomona, CA.